CLPS 1291 Final Project Report

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§1. Abstract

The goal of our project was to apply the deep learning and computational neuroscience skills we have learned in this class and apply them to the modality of audio. We use data from Nakai et al.[1] who took fMRI brain scans of five subjects as they listened to songs from the gtzan dataset. Gtzan contains songs from 10 genres with 100 songs for each genre.[2] We then reimplement the process described by Denk et al. in their paper *Brain2Music*[3] where they take these fMRI brain scans and reconstruct the song that the subject was listening to while the neural data was recorded. We provide some samples of original recordings and reconstructions that we created.

§2. Methods

The core model used by *Brain2Music* is MusicLM,[4] a model also developed by Agostinelli and Denk et al., which inputs a text prompt and outputs a generated song based on the semantic meaning of the text. In order to create *Brain2Music* it was necessary to first train MusicLM. An open source implementation[5] of MusicLM was created by Phil Wang and this was adapted to be used for our project. Wang's implementation includes the necessary architecture but lacks any pretrained models. As a result, we attempted to train this large model using the Music Caption Dataset for *Brain2Music*.[6] This dataset is composed of 540 song snippets each which has a corresponding text description. These songs line up with the same songs listened to by the participants in the fMRI machine. These song-text pairs were necessary to train MusicLM. MusicLM is comprised of two components. The first component is the autoencoder MuLan, which is composed of an audio and text transformer. During

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[1] Tomoya Nakai, Naoko Koide-Majima, and Shinji Nishimoto, Music Genre fMRI Dataset, 2021, OpenNeuro, [Dataset] doi: 10.18112/openneuro.ds003720.v1.0.0, https://openneuro.org/datasets/ds003720/.

[2] George Tzanetakis and Perry Cook, "Musical Genre Classification of Audio Signals," IEEE TRANSACTIONS ON SPEECH AND AUDIO PROCESSING 10, no. 5 (July 2002): 293–302, https://www.cs.cmu.edu/~gtzan/work/pubs/tsap02gtzan.pdf.

[3] Timo I. Denk, Yu Takagi, Takuya Matsuyama, Andrea Agostinelli, Tomoya Nakai, Christian Frank, and Shinji Nishimoto, "Brain2Music: Reconstructing Music from Human Brain Activity," 2023, arXiv, eprint 2307.11078, https://arxiv.org/pdf/2307.11078.pdf.

[4] Andrea Agostinelli, Timo I. Denk, Zalan Borsos, Jesse Engel, Mauro Verzetti, Antoine Caillon, Qingqing Huang, Aren Jansen, Adam Roberts, Marco Tagliasacchi, Matt Sharifi, Neil Zeghidour, and Christian Frank, "MusicLM: Generating music from text," 2023, arXiv, eprint 2301.11325, https://arxiv.org/abs/2301.11325.pdf

[5] Phil Wang, musiclm-pytorch, 2023, GitHub, [Source code], https://github.com/lucidrains/musiclm-pytorch

[6] NishimotoLab, Music Caption Brain2Music Dataset, 2023, Kaggle, [Dataset] https://www.kaggle.com/datasets/nishimotolab/music-caption-brain2music

training, the model takes in a song and a caption the two transformers attempt to create identical embeddings from these two input modalities. This way, during evaluation, only the input text prompt is embedded and propagated forward in the model to generate the new song. The second component is AudioLM, the predecessor to MusicLM, which is comprised of three transformer networks: Semantic, Coase, and Fine, as well as two pretrained auto-encoders: wav2vec, and Soundstream. In all this means that within MusicLM there were four submodules that all needed training. We had hoped to easily get MusicLM working in order to focus on its adaptation to *Brain2Music*. However, given the complexity and intricacies of MusicLM, we were forced to dedicate much more of our efforts to this portion of the project.

For AudioLM, the three transformers: Semantic, Coarse, and Fine were each trained for 1000 epochs and checkpoints were saved in order to load these models in for future use. However, for the fine transformer, we only used the checkpoint saved after 1 epoch because the loss quickly became nan after a few epochs. Mulan was also trained on multiple epochs but loss quickly plateaued. All of these models were trained using the Music Caption *Brain2Music* Dataset and the corresponding 540 songs in gtzan. This is a clear limitation as the original MusicLM was trained on around 50 million samples. We trained these networks using scripts[7] we wrote in Google Colab, in order to utilize the GPU compute offered.

Lastly, in order to replicate *Brain2Music*, the last step was to create the Linear Regression model which converts the fMRI scans into embeddings that can be used to pass into the music generating model(s). This was done by first passing all of the songs through MuLan in order to get the music embeddings. Then, each fMRI reading was paired with the correct Mulan song embedding of the song listened to during the scan. These pairing acted as the input-groundtruth pairs used to train our linear regression model, *W*, which was a single fully-connected linear layer with L2 regularization of weights. It was trained to reproduce the MuLan song embedding using only the fMRI data recorded when listening to that song. Since every brain is different, the model necessary to replicate the MuLan embeddings will be unique for every subject. As a result, we focused on training the linear regression for only one subject: Subject #1. We trained this network for 300 epochs. Then after training, We then apply *W* to novel fMRI data which was not used in the model's training. This generates music embeddings from the song which can then be passed into MusicLM in order to reconstruct the listened song. We trained on 50 songs and reconstructed 10. The generated 10 and the original songs they were trying to reconstruct are linked in the Results section.

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[7] Evan Li, Nick Masi, Theo McArn, MusicLM.ipynb, 2023, Google Colab, [Source code] https://colab.research.google.com/drive/1PxH4eZvd0RiwKpdPI1YXdP1Uz8T9hnUT?usp=sharing

§3. Challenges

One of our first challenges was with preprocessing the fMRI data. Since none of us had previous experience working with fMRI data, we found it difficult to recreate the original preprocessing steps of *Brain2Music*. We ended up following the preprocessing steps from Kunkhe et al[8], which utilizes SPM12 to apply motion correction, distortion correction, slice timing correction, coregistration of functional and structural images, normalization, and smoothing. Additionally, we could not figure out in a timely manner on how to find regions of interest (ROI) in fMRI data. However, for future work, by isolating specific regions of interest within our fMRI data, this would greatly reduce the dimensionality of our input data and would remove extraneous features. This would reduce complexity and increase the performance of our model.

Another major issue was the fact that we often ran out of memory on Colab. This occurred when preprocessing or training on the high dimensional neural data and/or the large music files. This forced us to use limited amounts of data: less of gtzan than we could have and less of the neural data than was made available by Nakai et al. Additionally, and related to the previous issue, we suspect that the ROI's found and used in the neural data serve as hand-selected features which enable the authors to effectively utilize a simple linear regression for *W*. We attempted to compensate for the fact that we didn't use this by using a deeper MLP model with non-linearities for *W*, but even for a relatively simple 3-layer model with ReLU in between our Colab sessions would run out of memory and crash. As a result we used the same linear regression approach as Denk et al.

The *Brain2Music* paper was unclear on their implementation at many points, especially when discussing the linear regression they used to transform fMRI data into the MuLan music embedding space. All of this data was very high dimensional because of the 4D nature of fMRI data (3D + time) plus the additional dimensions for which test run and on what subject. This made the situation all the more confusing. What we have ultimately determined that the paper does is that for each song a subject listens to, which are 15s long, the fMRI data is recorded in 10 1.5s intervals. The songs however are split into 4 10s intervals which are then embedded with MuLan. To match the dimensions so that a single matrix transformation can go from fMRI data to song embeddings (learning this transformation is training the linear regression *W*), they downsample the fMRI data by averaging sets of seven (though the paper they say five for some reason) 1.5s intervals to get 10.5s intervals. There are 4 such continuous sets in each 15s interval, and these 4 averaged sets of fMRI data are matched to the 4 embeddings per song for the purpose of training *W*. This was unclear to us at first, so we use the same window size of 1.5s for song embeddings as we do for fMRI data, so

[8] Philipp Kuhnke, Markus Kiefer, Gesa Hartwigsen, Task-Dependent Recruitment of Modality-Specific and Multimodal Regions during Conceptual Processing, Cerebral Cortex, Volume 30, Issue 7, July 2020, Pages 3938–3959, https://doi.org/10.1093/cercor/bhaa010

we directly correspond unaveraged fMRI scan windows to the same time window in songs. This yields 10 windows per song (this is variable *r* as defined in Table 1 of §3.3 of the paper), rather than 4. Each window is mapped to a 128-dimensional embedding by our trained *W*, and the average of these 10 embeddings are what is passed in to AudioLM to generate a song conditioned on the neural data.

This use of AudioLM is a solution to another issue we faced when attempting to reimplement the work of Denk et al. The MusicLM library we use from Wang does not allow for conditioning on music embeddings, only text. However, the AudioLM library does. Once we train the MuLan quantizer using MusicLM, we can pass it into AudioLM so that it behaves in essentially the same way as MusicLM. We thus create a workaround and do our audio generation/song reconstruction by passing in embeddings to AudioLM with the MuLan quantizer instead of MusicLM.

§4. Results

Below we provide a table of 10 songs listened to by subjects which we reconstructed from their fMRI data using our model.

Original Song	Blues	Classical	Country	Disco	Hiphop	Metal	Jazz	Pop	Reggae	Rock
Reconstructed Song	1	2	3	4	5	6	7	8	9	10

§5. Conclusion

Our original research idea relied on the fact that we thought with *Brain2Music* that the MusicLM model was learning from the fMRI data. If this were the case, then the final transformer might potentially have some more humanistic understanding of songs, similar to Harmonization[9] for vision. However, it turns out that the linear regression they use to transform the fMRI data into the song embedding space (which is then passed into MusicLM) is the only place humanistic vision might be gained, as the MusicLM model is not trained with any neural data and is then frozen to use for inference. The linear regression uses the MuLan embeddings for songs as the ground truth, so MusicLM is given something in the shape of MuLan embeddings it is familiar with. This allows novel neural data at inference time to be mapped to the MuLan latent space, and then passes this into MusicLM (which has been trained in its standard manner with no neural data) so that a song can be generated. This means the novel addition of

[9] Thomas Fel, Ivan Felipe, Drew Linsley, and Thomas Serre, “Harmonizing the object recognition strategies of deep neural networks with humans,” *Advances in Neural Information Processing Systems (NeurIPS)*, 2022, https://github.com/serre-lab/Harmonization.

Brain2Music was the simple linear regression model which learns a mapping from neural data to the MuLan embedding space, not anything to do with MusicLM! Unfortunately, it also meant there is likely no encoded human knowledge of music in MusicLM, though the MuLan embeddings might have some new human flavor to them. If that is the case, it would imply that the humanistic understanding of audio does exist in the linear regression model *W*.

One piece of future research we are very interested in is whether a linear probe can be added to *W* in order to perform music genre recognition (MGR) on the inputted song. This would investigate whether the model learning from fMRI data on listening to songs improves the model's performance on downstream audio tasks. Furthermore, because training of our *W* was on such limited song and neural data, it would be interesting to see the impact of greater scale (more data and more advanced models, such as a transformer instead of linear regression) on MGR or other downstream audio tasks.

Overall, we were not able to reconstruct songs at high fidelity nor accurately. This is due mostly to limited amounts of training data, and the lack of time and compute to even process and utilize all of the training data that we did have available. However, the three of us learned a tremendous amount about the state of the art of song generation, both at a theoretical level and with the practical implementation, as well as an innovative way to apply computational neuroscience techniques to the deep learning domain. We also developed our skills in the whole processing pipeline—from preprocessing to use in training models—of neural data which was a different modality (fMRI) than anything covered in class. Furthermore, we are very excited that we were able to generate songs at all, as this was a novel form of generative model for all of us.

Code and Music Repository

Update - Tuesday, December 5

So far we have accomplished all three of our original tasks. Chiefly, we have obtained both the fMRI data from Denk et al. and the gtzan song dataset. The preprocessing of the fMRI data took longer than expected as the dataset was complex and it required experimentation with different programs. We were eventually able to apply realignment & unwarping, skull stripping, normalization, and smoothing to preprocess the data. However, we were unable to completely replicate the original paper's preprocessing steps due to time constraints. We are a little behind schedule with some of our other subtasks. Namely, MusicLM is more complicated to get working than we expected. MusicLM utilizes both the MuLan multimodal contrastive model as well as AudioLM, which itself is composed of three different transformers. All of these models must be pretrained before they can be composed and used for inference as MusicLM. However, we still believe it may be possible for us to complete Phase 1 by this Thursday, and will be meeting Wednesday and Thursday to work on this. Doing so would involve recreating the song reconstruction results from Denk et al. If we are not able to accomplish this by this week, then we may drop our second project goal (benchmarking MusicLM on MGR, which was Phase 2) and just focus on reconstructing MusicLM and the results from Denk et al. We discussed with Prof. Serre and believe that doing Phase 1 alone would be a suitable project because it deals with many concepts from the class such as neural data and advanced transformer-based models. Once again though, we will attempt to achieve both goals still. We have started getting the components of MusicLM setup and will continue this work when we meet Wednesday and Thursday. Additionally, the process of transforming fMRI data into embeddings which can be passed into MusicLM is more complex than originally anticipated and is bespoke for each different person's neural activity, so we will only train the model to do this conversion for one patient and attempt to reconstruct the songs they listened too. Figuring out how to recreate the transformation process will take considerable work. We have read the section in the paper that covers this (which is not written in great detail) and started thinking about how to do it, and will begin writing code to implement it tomorrow.

CLPS 1291 Final Project Proposal

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For our final project we propose working with audio as a modality. Most of this course has focused on visual stimuli, we thought that it would be interesting to apply traditional deep learning techniques to a different modality we have less experience with. To bridge our interest in music with the computational neuroscience methods we have learned in this course, we want to see how deep learning can be used to reconstruct music solely by understanding the representations of music in human brain activity. The motivation behind this task is that training this model can help us understand more about how the brain stores and processes auditory stimuli and how the brain interprets and distinguishes songs. This may result in neural networks that understand audio as a modality in a manner more aligned with how humans do; making the work similar to what Harmonization has done for visual models.

We have identified the paper *Brain2Music: Reconstructing Music from Human Brain Activity* by Denk et al. as among the state of the art in the field of music reconstruction from neural data.[10] This work builds on a previous paper out of Google from Agostinelli, Denk, et al. where they introduce the MusicLM model which takes text input as prompts and generates songs.[11] The *Brain2Music* paper utilizes linear regression to modify fMRI data such that it can be passed into MusicLM. The output is a reconstruction of the song being listened to as the neural data was recorded. For our work, we plan to replicate the model proposed in the paper to see if we can get similar results ourselves. Furthermore, we want to see if leveraging this network trained on neural data, we can develop a model that is better at the music genre recognition (MGR) task than other models which are not exposed to any neural data.

Other work which we consider more rudimentary and less interesting was done by Bellier et al. who reconstructed a Pink Floyd song from neural data but did so with a simple multi-layer perceptron.[12] Additionally, it uses only one song which we intuit will not lend well towards transferring to the MGR task. OpenAI also has a model MuseNet[13] that creates songs, but we do not believe it is within scope for us to attempt to recreate their work nor does it have any tie into cognitive science or neural data. Also, the paper

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[10] Timo I. Denk, Yu Takagi, Takuya Matsuyama, Andrea Agostinelli, Tomoya Nakai, Christian Frank, and Shinji Nishimoto, "Brain2Music: Reconstructing Music from Human Brain Activity," 2023, arXiv, eprint 2307.11078, https://arxiv.org/pdf/2307.11078.pdf.

[11] Andrea Agostinelli, Timo I. Denk, Zalán Borsos, Jesse Engel, Mauro Verzetti, Antoine Caillon, Qingqing Huang, Aren Jansen, Adam Roberts, Marco Tagliasacchi, Matt Sharifi, Neil Zeghidour, Christian Frank, "MusicLM: Generating Music from Text," 2023, arXiv, eprint 2301.11325, https://arxiv.org/pdf/2301.11325.pdf.

[12] Bellier L, Llorens A, Marciano D, Gunduz A, Schalk G, Brunner P, Knight RT, "Music can be reconstructed from human auditory cortex activity using nonlinear decoding models," PLoS Biol, 2023 Aug 15; 21(8):e3002176, doi: 10.1371/journal.pbio.3002176, PMID: 37582062; PMCID: PMC10427021, https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10427021/.

[13] Payne, Christine. "MuseNet." *OpenAI*, 25 Apr. 2019, openai.com/blog/musenet.

and model architecture are not in the public domain so it is even less possible for us to recreate.

Denk et al. reconstructed songs from the GTZAN dataset[14] which is primarily used for the MGR task. This solidifies our interest in recreating their work because we then hope to transfer the audio-reconstruction network to the classification task using the same GTZAN dataset. This presents a good opportunity to build on top of the work from Denk et al. because the dataset they use for reconstruction is among the most popular for the task of MGR[15] with a leaderboard available on PapersWithCode.[16] The idea is that we can study the effect that learning from human's neural representation of audio has on the ability for the model to classify human-defined genres. Specifically, after training the MusicLM model on the song reconstruction task, we will replace the linear regression model at the input to fit the GTZAN data, and then at the output we will attach a new output classification model to receive our genre classification. In doing so, we can now pass in songs (from GTZAN) into the same MusicLM model as before, and use the new classifier at the end to output the predicted genre. We anticipate challenges from using music as a modality because it is not something that any of us have experience with.

Our plan includes Phase 1: finishing the reimplementation of results from Denk et al. by Thursday, December 7th. This will yield a model that can interpret human neural data from listening to songs. We will also write up our progress, challenges, and results of this first phase as we go. Over the next week, until the deadline on Thursday, December 14th, we will work on Phase 2 where we will attempt to transfer this model to the genre classification task posed by the GTZAN dataset. A rough outline, subject to changes as we actually proceed with work, for the division of labor within the project is as follows:

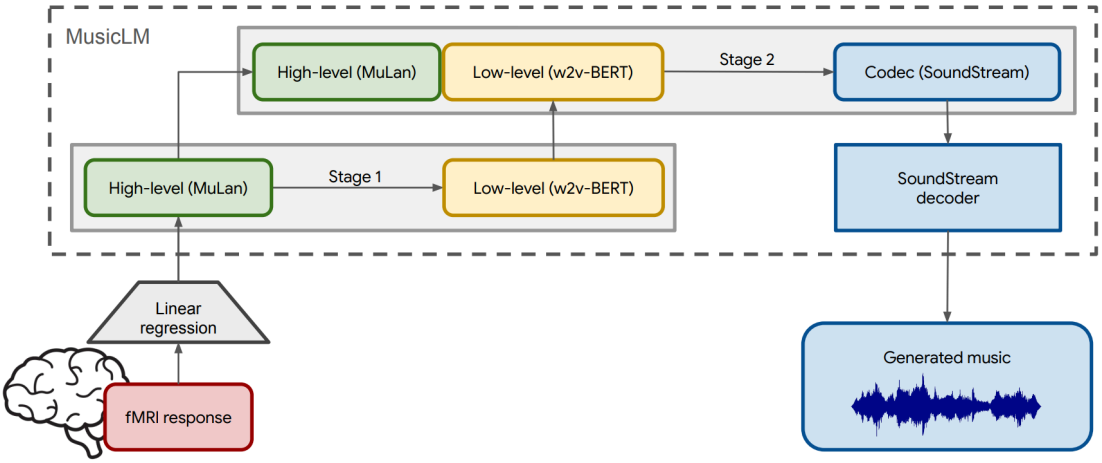
Tasks	Assigned	Complete by
Project proposal	Nick & Theo	11/28
Downloading/preprocessing the GTZAN dataset	Nick	12/5
Downloading/preprocessing the fMRI data from Denk et al.	Evan	12/3

[14] George Tzanetakis and Perry Cook, "Musical Genre Classification of Audio Signals," *IEEE TRANSACTIONS ON SPEECH AND AUDIO PROCESSING* 10, no. 5 (July 2002): 293–302, https://www.cs.cmu.edu/~gtzan/work/pubs/tsap02gtzan.pdf.
[15] Bob L. Strum, "The GTZAN dataset: Its contents, its faults, their effects on evaluation, and its future use," 2023, arXiv, eprint 1306.1461, https://arxiv.org/pdf/1306.1461.pdf.
[16] https://paperswithcode.com/dataset/gtzan.

Getting Brain2Music model (MusicLM) working from Denk et al. working	Theo & Evan	12/3
Recreate results	All	12/5
Write up on Phase 1	All	12/7
Train a new linear regression and classification model at the input & output of MusicLM to take song input and output a genre classification; then benchmark on gtzan	All	12/12
Final write-up	All	12/14

Working References/Links:

- GTZAN dataset
    - Tensorflow
    - PyTorch
- fMri Data
- Google neural reconstruction of GTZAN songs (simplified?)



- Linear Regression
    - Adjusted based on the person
- MusicLM (open source implementation, pytorch)
    - Components:
        - Wav2vec (w2v)-BERT
        - MuLAN
        - SoundStream
    - Caption data
    - Neural data
- Pink Floyd song reconstruction with MLP
- Song compression (maybe some room for feature selection, preprocessing that would affect reconstruction quality)
- OpenAi MuseNet
- How to Guide sort of
    - Music temporal → RNN
    - Multiple channels
- GTZAN preprocessing